

EDUCATION

Carnegie Mellon University

Ph.D Candidate in Mechanical Engineering - Robotics

M.S. in Mechanical Engineering (Research) - Robotics and Control Systems

Massachusetts Institute of Technology

B.S. in Mechanical Engineering, Electrical Engineering & Computer Science

Pittsburgh, PA

May 2024 – May 2028

Aug 2022 – May 2024

Cambridge, MA

Sept 2018 – May 2022

SKILLS

Languages: Python, C++, C, MATLAB, Julia, Assembly, HTML, CSS, Bash, CMake

Software: ROS1/2, IsaacSim/Lab, Gazebo, Mujoco, RViz, Docker, AWS, GCP, OpenCV, Tensorflow, Pytorch

Areas of Expertise: Trajectory Optimization, Optimal Control, Reinforcement Learning, Multi-Robot Motion Planning, State Estimation

RESEARCH EXPERIENCE

CMU Robomechanics Lab

Graduate Research Student

Pittsburgh, PA

Sept 2022 – Present

- Maintainer of Quad-SDK, a widely adopted open-source, full-stack software framework for quadrupedal locomotion with 900+ GitHub stars.
- Engineered and deployed learned locomotion and disentanglement policies for quadrupedal robots navigating dense compliant vegetation, improving robustness and recovery in contact rich natural environments.
- Designed autonomous whole-body control frameworks for terrain-modifying robots (digging, excavation, plowing) that leverage terra-mechanics-based ground interaction models into training, enabling stable and efficient execution across varying soil conditions.
- Implemented multi-agent coordination algorithms for heterogeneous robotic teams enabling safe, real time operation near humans, obstacles and other agents.
- Developing adaptable locomotion policies for robots traversing deep mud by addressing complex physical phenomena, including suction effects, non-Newtonian material behavior, and material accumulation.

PROJECTS

Multi-Robot Motion Planning for Quadruped Robots

Carnegie Mellon - 16.782

Pittsburgh, PA

Sept 2023 – Dec 2023

- Integrated three algorithms to facilitate multi-robot motion planning of quadruped robots in C++; namely a Sequential RRT-Connect, Joint Space RRT-Connect and Conflict Based Search.
- Conducted a performance evaluation study, revealing that the CBS exhibited better scalability beyond 4 robots, demonstrating the fastest planning time, and achieving an average path length 17.5% and 10.3% shorter than those of the Sequential and Joint methods respectively.

Hybrid Trajectory Optimization for a Monkey Bar Robot using DIRCOL

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Pittsburgh, PA

Jan 2023 – May 2023

- Created a hybrid-system direct collocation (DIRCOL) trajectory optimization-based controller of a two-link Monkey-Bar robot in Julia. Utilized IPOPT solver and MeshCat libraries for visualization.

PUBLICATIONS & ABSTRACTS

Proprioception and Reaction for Walking Among Entanglements

Justin K. Yim, Jiming Ren, **David Ologan**, Selvin Garcia Gonzalez, and Aaron M. Johnson

In IEEE/RSJ International Conference on Intelligent Robots and Systems, October 2023

Quad-SDK Update: Estimation, Underbrush, and Other Improvements

David Ologan, Ardalan Tajbakhsh, Justin K. Yim, Yanhao Yang, Joseph Norby, Jiming Ren, Selvin Orlando Garcia Gonzalez, and Aaron M. Johnson

In IROS Late Breaking Results, October 2023